

HW 11

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1. What makes the problem of object detection and recognition different than image classification?

Image classification simply assigns a label to the entire image without specifying object positions. Detection involves not only identifying what objects are present in an image but also locating them by drawing bounding boxes around each instance. So, it requires handling multiple objects, varying scales, and overlapping instances, making it more complex than classification, which assumes a single dominant object per image.

2. How does the Faster R-CNN network decide what parts of the image to analyze?

Faster R-CNN uses a Region Proposal Network (RPN) to efficiently determine which parts of the image to analyze. The RPN slides over the convolutional feature map and predicts potential regions (called "anchors") that likely contain objects, making it faster and more accurate.

3. How do the YOLO style networks decide what parts of the image to analyze?

YOLO divides the input image into a grid, and each grid cell predicts bounding boxes and class probabilities directly. Unlike Faster R-CNN, YOLO does not use region proposals—instead, it processes the entire image in a single pass, making it extremely fast. Each grid cell is responsible for detecting objects whose centers fall within it, and the network predicts multiple boxes with confidence scores, keeping only the most likely detections.

4. When doing transfer learning, what part of the deep network is modified to adapt to the new problem?

In transfer learning, the pre-trained backbone (like ResNet or VGG) is typically kept frozen to preserve learned features, while the final fully connected layers (the "head" of the network) are replaced and retrained for the new task. This allows the model to leverage general feature extraction from the original dataset while adapting the last layers to the specific classes or detection requirements of the new problem.

5. What is the purpose of the deep network when doing a matching problem, such as face recognition?

The deep network's purpose is to generate an embedding—a compact, discriminative feature vector that represents the input (e.g., a face). The network is trained so that embeddings of similar faces are close together in the feature space, while dissimilar faces are far apart. The actual matching is done by comparing these embeddings (e.g., using cosine similarity) rather than directly classifying identities.

6. What methods are used to do the actual matching of an input to a database of examples?
 - Cosine Similarity: Measures the angle between two vectors.
 - Euclidean Distance: Computes the straight-line distance between embeddings.
 - k-Nearest Neighbors (k-NN): Finds the closest matches in the database. For large-scale systems, approximate nearest neighbor search techniques like FAISS or Annoy are used to speed up retrieval.